METRIC

MIL-PRF-24574A(SH)
31 March 2025
SUPERSEDING
DOD-PRF-24574(SH)
29 September 1978

PERFORMANCE SPECIFICATION

LUBRICATING FLUID FOR LOW AND HIGH PRESSURE OXIDIZING GAS SYSTEMS

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lubricating fluid for use in low and high pressure oxidizing gas systems, including oxygen and compressed air. The lubricating fluid is intended for use with oxygen systems at all pressures, and for air and oil-free nitrogen systems at pressures above 10,300 kilopascals (kPa) (1,500 pounds per square inch [psi]).

1.2 Classification. Based on viscosity, volatility, and other physical properties, the lubricating fluid is of the following types, as specified (see 6.1 and 6.2):

1.2.1 Types.

a. Type I – for use at temperatures from -46 to 0 °C (-50 to 32 °F) (see 6.1.1).

b. Type II – for use at temperatures from -20 to 40 °C (-4 to 104 °F) (see 6.1.2).

c. Type III – for use at temperatures from 20 to 70 °C (68 to 158 °F) (see 6.1.3).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications forms a part of this specification document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9150

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise indicated, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D92 - Standard Test Method for Flash and Fire Points by Cleveland Open Cup

ASTM D97 - Standard Test Method for Pour Point of Petroleum Products

ASTM D130 - Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test

ASTM D287 - Standard Test Method for API Gravity of Crude Petroleum and Petroleum Products (Hydrometer/Method)

ASTM D445 - Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)

ASTM D665 - Standard Test Method for Rust-Preventing Characteristics of Inhibited Mineral Oil in the Presence of Water

ASTM D892 - Standard Test Method for Foaming Characteristics of Lubricating Oils

ASTM D974 - Standard Test Method for Acid and Base Number by Color-Indicator Titration

ASTM D1160 - Standard Test Method for Distillation of Petroleum Products at Reduced Pressure

ASTM D1218 - Standard Test Method for Refractive Index and Refractive Dispersion of Hydrocarbon Liquids

ASTM D1500 - Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale)

ASTM D2619 - Standard Test Method for Hydrolytic Stability of Hydraulic Fluids (Beverage Bottle Method)

ASTM D4057 - Standard Practice for Manual Sampling of Petroleum and Petroleum Products

ASTM D4066 - Standard Classification System for Nylon Injection and Extrusion Materials (PA)

ASTM D4172 - Standard Test Method for Wear Preventive Characteristics of Lubricating Fluid (Four-Ball Method)

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS7276 - Rings, Sealing, Fluorocarbon Rubber High-Temperature-Fluid-Resistant, Very-Low-Compression-Set FKM Type 70 – 80

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

SAE AMS3216 - Fluorocarbon Rubber Fuel and Oil Resistant 70 – 80

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Composition. The composition of these fluids is not specified. However, in order to meet the ignition temperature test (see appendix A) and other requirements, it shall be necessary to provide fully halogenated products in which all the hydrogen has been replaced with fluorine, or fluorine and chlorine. It shall also be necessary to have at least three times as many fluorine atoms as chlorine atoms in the polymer unit molecular structure. These fluids shall be of the generic type chlorotrifluoroethylene polymer, or be fully compatible and soluble with the chlorotrifluoroethylene polymers.

3.3 Physical and chemical characteristics. Lubricating fluids shall conform to the requirements of table I for the type specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

TABLE I. Physical and chemical characteristics.

Property	Requirement			Verification
	Type I	Type II	Type III	
Kinematic viscosity:				
@ 100 °C (212 °F), mm^2/s (cSt)	1.0 to 2.0	2.5 to 4.0	12.0 to 20.0	ASTM D445
@ 40 °C (104 °F), mm^2/s (cSt)	4.0 to 5.0	13 to 35	100 to 450	
@ -20 °C (-4 °F), mm^2/s (cSt)	110 max.	---	---	
Pour point, °C (°F), max.	-62 (-80)	-20 (-4)	0 (32)	ASTM D97
Distillation at reduced pressure				
Vapor temperature, 1 mm Hg pressure, 1 percent, °C (°F), min.	35 (95)	80 (176)	133 (268)	ASTM D1160
Copper strip corrosion test at 100 °C (212 °F), max.	1	1	1	ASTM D130
Rust protection	pass	pass	pass	ASTM D665, Proc C 1/
Flash point, °C (°F), min.	none	none	none	ASTM D92 2/
Fire point, °C (°F), min.	none	none	none	ASTM D92 2/
Autogenous ignition temperature in oxygen at 10,300 kPa (1,500 psi), °C (°F), min.	350 (662)	350 (662)	350 (662)	See appendix A and 4.4.1
Wear test, scar diameter, mm, max.	0.5	0.5	0.5	ASTM D4172 3/
Refractive index, nD at 20 °C (68°F)	report 4/	report 4/	report 4/	ASTM D1218 4/
Specific gravity at 15.6 °C (60 °F)	report 4/	report 4/	report 4/	ASTM D287 4/
Hydrolytic stability:				
Copper specimen wt loss, mg/cm^2	0.3	0.3	0.3	ASTM D2619
Fluid acid No. increase, mg KOH/g	0.2	0.2	0.2	
Water acid No. increase, mg KOH/g	5.0	5.0	5.0	
Insoluble, percent max.	0.5	0.5	0.5	
Color number, ASTM	report 4/	report 4/	report 4/	ASTM D1500 4/
Neutralization number, mg KOH/g, max.	0.5	0.5	0.5	ASTM D974
Compatibility	pass	pass	pass	See 4.4.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

TABLE I. Physical and chemical characteristics – Continued.

Property	Requirement			Verification
	Type I	Type II	Type III	
Foaming characteristics				ASTM D892
Sequence 1, mL, max.	300	300	300	
Sequence 2, mL, max.	25	25	25	
Sequence 3, mL, max.	300	300	300	
Toxicity	pass	pass	pass	See 4.4.3

NOTES:

1/ Use procedure C for fluids heavier than water.

2/ Use hood. Start testing at 150 °C (302 °F) or 30 °C (86 °F) below boiling, no flash or fire below 400 °C (752 °F).

3/ Load: 15 kg; oil temperature: 80 °C (176 °F); speed: 600 r/min; test duration: 120 minutes.

4/ No limiting requirement, test made for record.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

3.4 Toxicity. When evaluated in accordance with 4.4.3, the lubricating fluid shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.4.3 and 6.4).

3.5 Prohibited materials. The lubricating fluid shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs.

3.6 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections.

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on candidate oil when a first article sample is required (see 3.1). This inspection shall consist of the tests specified in table I. Failure of the material to pass any test shall be cause for rejection of the first article.

4.2.1 Sampling. Two 5-gallon all-level samples shall be taken from the storage tank or from two containers selected at random. The samples shall be placed in separate clean, dry, metal or glass containers, dyed as specified (see 6.2), and sealed, marked and forwarded to a laboratory satisfactory to the contracting activity for inspection.

4.3 Conformance inspection. Conformance inspection shall consist of the tests specified in table II, except that the corrosion, rust protection, wear, hydrolytic stability, compatibility, foaming characteristics, and autogenous ignition tests shall not be required.

TABLE II. First article and conformance tests.

Tests	First article	Conformance
Kinematic viscosity		
at 100 °C	X	X
at 40 °C	X	X
at -20 °C	X	X
Pour point	X	X
Distillation at reduced pressure		
Vapor temperature	X	X
Corrosion test at 100 °C (212 °F)	X	
Rust protection	X	
Flash point	X	X
Fire point	X	X
Autogenous ignition temperature in oxygen at 10,300 kPa (1,500 psi)	X	
Wear test, scar diameter	X	
Refractive index at 20 °C (68 °F)	X	X
Specific gravity at 15.6 °C (60 °F)	X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

TABLE II. First article and conformance tests - Continued.

Tests	First article	Conformance
Hydrolytic stability		
Copper specimen wt loss	X	
Fluid acid No. increase	X	
Water acid No. increase	X	
Insoluble	X	
Color number	X	X
Neutralization number	X	X
Foaming characteristics	X	
Toxicity	X	
Prohibited materials	X	

4.3.1 Sampling for conformance.

4.3.1.1 Inspection lot.

4.3.1.1.1 At place of manufacture. A lot of lubricating fluid shall consist of a manufacturer's batch. When material cannot be identified by batch, a lot shall consist of not more than 350 kilograms (kg) (800 pounds [lbs]), and shall be from a single mixing container.

4.3.1.1.2 At place of delivery. For the purpose of sampling, a lot of lubricating fluid shall consist of all the fluid in a single shipment.

4.3.1.2 Sampling a storage tank. A representative sample of 4 L (1 gallon) of lubricating fluid shall be taken from each lot in accordance with ASTM D4057.

4.3.1.3 Sampling during loading. Samples shall be drawn at the discharge pipe where it enters the bulk carriers (tank cars or tank trucks). At least four samples shall be taken at regular intervals during the entire period of loading or filling, each sample being 1 pint. Only one sample for one continuous drum filling operation is required and shall be taken after approximately 10 percent of the volume of product has been drummed.

4.3.1.4 Sampling of filled containers. Where the lubricating fluid is contained in containers larger than 1 L (1 quart), these containers shall be sampled by mixing thoroughly and withdrawing 1 L (1 quart) from at least five containers, or all the containers, if fewer than five. However, the total sample drawn shall be at least 4 L (1 gallon) for conformance testing.

4.3.1.5 Examination of filled containers. A random sample of filled containers shall be selected from each lot in accordance with ANSI/ASQ Z1.4 at inspection level I and Acceptable Quality Level (AQL) as specified (see 6.2). Samples shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample filled container shall also be weighed to determine the amount of the contents. Samples having one or more defects or under required fill shall be cause for rejection of the container. If the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of ANSI/ASQ Z1.4, this shall be cause for rejection of the lot represented by the sample.

4.3.2 Acceptance criteria.

4.3.2.1 Refractive index. The refractive index from lot to lot shall be within 0.005 units and 2.0 percent of the average.

4.3.2.2 Specific gravity. The specific gravity from lot to lot shall be within 0.005 units and 2.0 percent of the average.

4.3.3 Rejection of lots. If any sample of lubricating fluid fails any of the tests, this shall be cause for rejection of the lot represented by the sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

4.4 Test methods. Test procedures shall be in accordance with table I, with the following additions.

4.4.1 Autogenous ignition temperature. See appendix A.

4.4.2 Compatibility. Each manufacturer's product shall be mixed with three (or all, if less than three) other manufacturers' products in clean glass containers (50:50) and stored at room temperature at 60±3 °C (140±5 °F) for seven days. Any precipitate or cloud shall be evidence of incompatibility. Also, specimens of fluorocarbon rubber in accordance with SAE AMS 7276 or SAE AMS 3216, and specimens of polyamide in accordance with ASTM D4066, composition A, shall be immersed in portions of the sample and held at 60 °C (140 °F) for seven days. Softening or swelling of the specimen shall constitute evidence of incompatibility.

4.4.3 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.4 as specified (see 6.2). The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the lubricating fluid using data provided by the manufacturer/distributor to the NMCFHPC (see 3.4 and 6.4).

4.4.4 Prohibited materials. Prohibited materials shall be verified as specified (see 3.5 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The lubricating fluid is use in low and high pressure oxidizing gas systems, including oxygen and compressed air. Current stock of lubricating oil may be used until depleted; future requisitions for lubricating oil should refer to the current version of this specification. The lubricating fluid is intended for use with oxygen systems at all pressures, and for air and oil-free nitrogen systems at pressures above 10,300 kilopascals (kPa) (1,500 pounds per square inch [psi]).

6.1.1 Type I lubricating fluid. Type I lubricating fluid is intended for use where a moderate viscosity is required at operating temperatures from -46 °C to 0 °C (-50 to 32 °F), for example, as a crankcase lubricant in compressors for low temperature liquid air distillation plants.

6.1.2 Type II lubricating fluid. Type II lubricating fluid is intended for use where this viscosity is required at operating temperatures from -20 °C to 40 °C (-4 to 104 °F), for example, as a compressor lubricant within the prescribed temperature range, such as with liquid oxygen transfer pumps.

6.1.3 Type III lubricating fluid. Type III lubricating fluid is intended for use where this viscosity is required at operating temperatures from 20 °C to 70 °C (68 to 158 °F), as a crankcase lubricant in nominal temperatures within the prescribed temperature range.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of fluid required (see 1.2 and 6.1).
c. When first article inspection is required (see 3.1).
d. Requirements for an HHA (see 3.4 and 4.4.3).
e. Requirement for prohibited materials verification (see 3.5 and 4.4.4).
f. AQL of 1.0 (see 4.3.1.5).
g. Level of packaging and packing required (see 5.1).
h. Activities requiring copies of completed Safety Data Sheets (SDS) (see 6.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

i. Quantity. The material should be purchased by volume, the unit being 3.785 L (1 gallon) at 15.6 °C (60 °F), or a specific number of standard size containers (see MIL-STD-290).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should be provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume I, Shelf-Life Management Program: Program Administration; and Volume 2, DoD Shelf-Life Management Program: Materiel Quality Control Storage Standards. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil.

6.5 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

6.6 SDSs. Contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313, as specified (see 6.2). In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.7 Subject term (key word) listing.

Acetone
Chlorotrifluoroethylene
Chromic acid
Compressed air
Trichlorotrifluoroethane

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

DETERMINATION OF AUTOGENOUS IGNITION TEMPERATURE AT HIGH PRESSURE

A.1 SCOPE

A.1.1 Scope. This appendix specifies the procedure for determining the high pressure autogenous ignition temperature (AIT) of lubricating fluids for low and high pressure oxidizing gas systems. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

A.2.2 Non-Government documents.

COMPRESSED GAS ASSOCIATION (CGA)

CGA G-4 - Cleaning of Equipment for Oxygen Service

(Copies of this document are available online at https://www.cganet.com/.)

A.2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

A.3 EQUIPMENT

A.3.1 Assembly. Suitable components shall be assembled so that the specified reaction vessel (bomb), including sample holding assembly, can be charged with oxygen and heated, and a record kept of the time and temperature at which ignition occurs. One possible assembly is shown on figure A-1.

A.3.2 Reaction vessel. The reaction vessel shall be the American Instrument Company (AMINCO) Model No. 41-12555, or equal.

A.3.3 Thermocouple assembly. The thermocouple assembly shall be AMINCO No. 45-1760, or equal.

A.3.4 Sample holding assembly. The sample holding assembly shall be as shown on figure A-2.

A.3.5 Test tube stopper. The test tube stopper shall be fabricated from the borosilicate glass tubing of such a size that it just fits inside the larger test tube, as shown on figure A-3.

A.3.6 Supporting spring. The supporting spring shall be wound from B and S 22-24 gauge nickel-chromium resistance wire by forming several turns on a mandrel of such a size that the resulting spring fits snugly over the larger test tube.

A.3.7 Support loop. The support loop shall be formed to hold the test tube at the proper height in the bomb; for example, the sample and thermocouple tip are roughly equidistant from the center of the zone heated by the heating jacket.

A.3.8 Thermocouple tip. The thermocouple tip shall be 15±2millimeters (mm) inside the top of the small, inner test tube. The entire assembly shall be hung from the bomb cover by a bushing in the bomb cover.

A.4 PROCEDURE

a. Clean all components of the assembly. Glass parts should be soaked in chromic acid cleaning solution, rinsed in distilled water, and dried. Metal components should be clean and bright. Metal components shall be rinsed in the trichlorotrifluoroethane or acetone, polished with steel wool, and re-rinsed with solvents. However, all traces of steel wool shall be eliminated since it is highly flammable.

b. Weigh out a 0.20±0.03 gram (g) sample into the inner culture tube.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

c. Assemble the equipment as shown on figures A-1 and A-2, and as directed by the bomb manufacturer.

d. Flush the bomb twice with oxygen meeting the requirements of CGA G-4, by pressurizing to 5,000 kPa (725 psi), and releasing the pressure.

e. Fill the bomb with the above specified oxygen to a pressure of 11,500 kPa (1,650 psi) and allow to stand at room temperature for 15 minutes. The pressure will drop about 300 kPa (45 psi) during the first 5 minutes while the gas cools, but it shall be nearly constant thereafter. Any additional steady pressure drop indicates a leak in the system that shall be corrected before proceeding. After assuring the absence of leaks, the pressure shall be adjusted to 10,300 kPa (1,500 psi).

f. Heat the bomb at a rate of 5±1 °C (9±2 °F) per minute. It will take a while for the heating rate to become steady, but the specified rate should be maintained from 60 °C to 260 °C (140±500 °F). Above 260 °C (500 °F), it may be difficult to maintain this heating rate, but it shall not drop below 3 °C (5 °F) per minute.

g. Ignition of the sample shall be indicated by a very rapid temperature rise of at least 20 °C (36 °F). As soon as ignition is complete, but not less than 3 minutes after it starts, turn off the heater and recorder, and release the bomb pressure under a laboratory-type hood, some decomposition products are corrosive, irritating, or even poisonous.

h. If no ignition occurs below the maximum safe operating temperature of the bomb 425 °C (800 °F), stop heating and report as above 425 °C (800 °F).

i. Report the sample size, initial and final gas pressure, and the ignition temperature.

A.5 LIMITATIONS

Occasionally, the decomposition and oxidation of fully fluorinated materials releases so little energy that there is no clear-cut indication of ignition. Also, if the sample distills into other parts of the bomb before the ignition temperature is reached, there shall be no clear indication of ignition. Neither of these situations shall arise with the chlorotrifluoroethylene polymers desired under this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

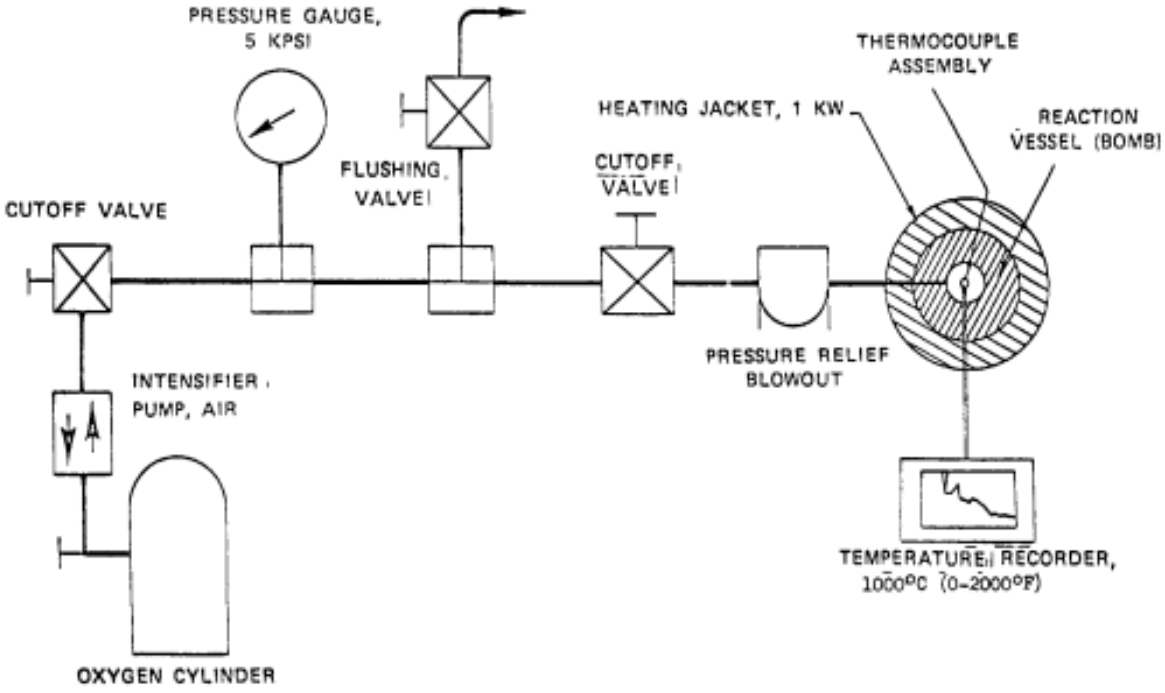


FIGURE A-1. AIT equipment assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

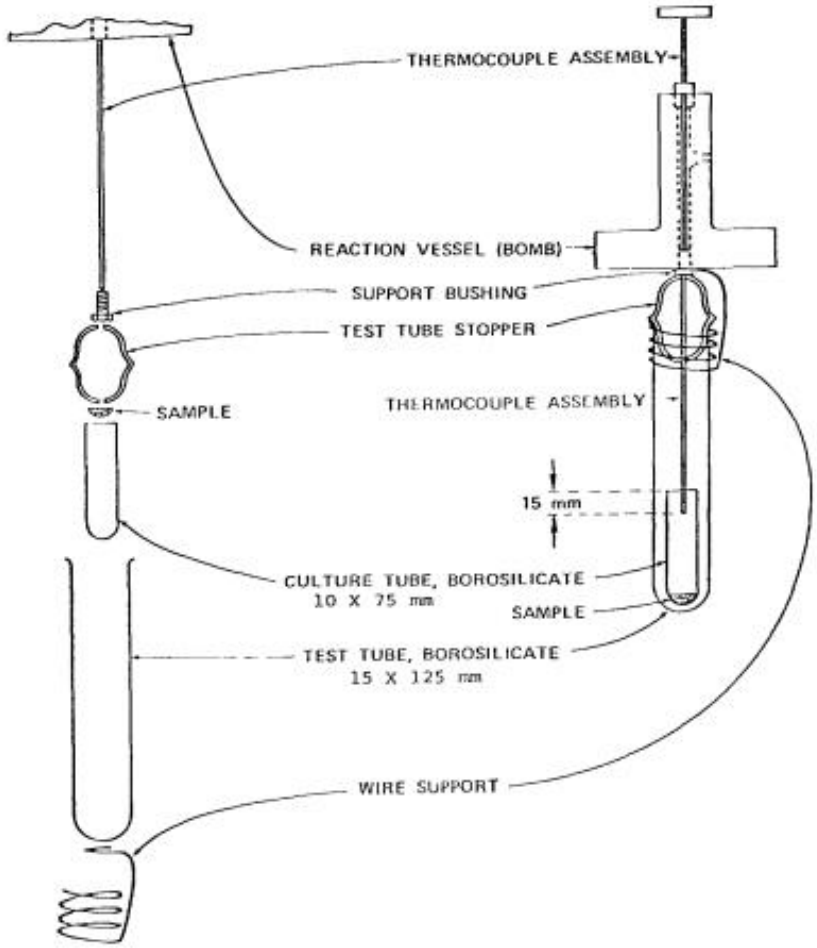


FIGURE A-2. Sample holding assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

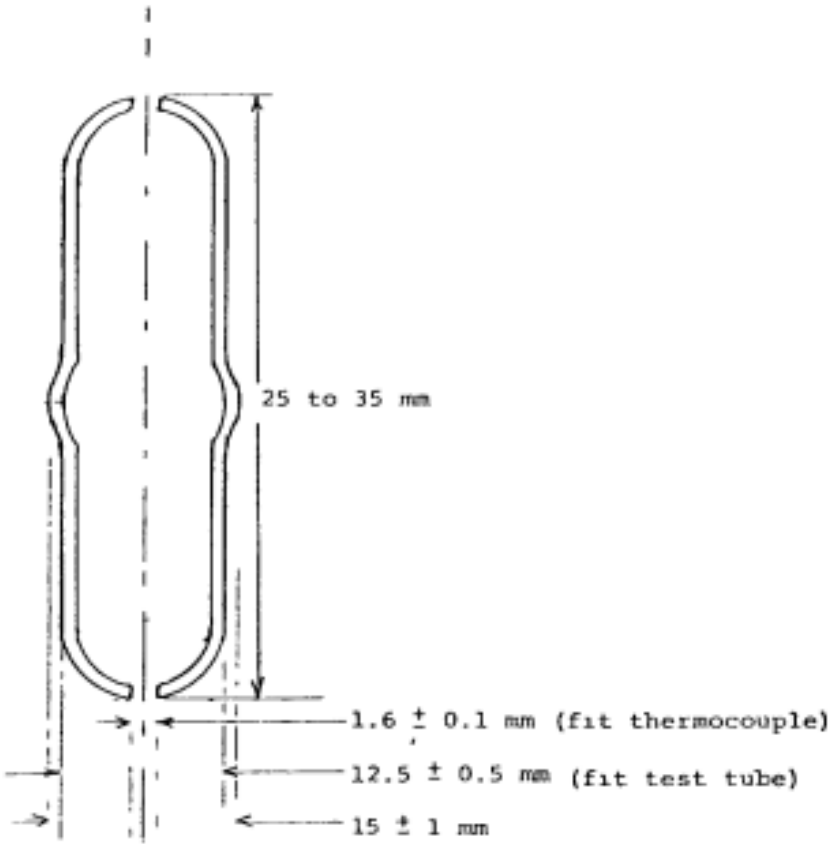


FIGURE A-3. Test tube stopper.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy – SH
(Project 9150-2024-004)

Review activities:
Air Force – 68
DLA – GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:04Z
Check the source to verify that this is the current version before use.